

Technology Skill Development Lab Java

Assignment No.2

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Problem Statement : Design a class named "Product" with default, and parameterized constructors. a. Default constructor initializes default values b. the parameterized constructor allows custom initialization Implement methods to display product details. Test the class by creating instances using each constructor to make sure everything works correctly and data is handled properly. creating multiple product objects, store their data in a suitable collection (such as an ArrayList) and perform basic data analysis on the product dataset.

Program:

```
import java.util.ArrayList;


class Product {


    int product_id;

    String product_name;

    double price;


    Product() {

        product_id = 0;

        product_name = "Default";

        price = 0.0;

    }


    Product(int id, String name, double pr) {

        product_id = id;

        product_name = name;
```

```
        price = pr;
    }

    void display() {
        System.out.println("Product ID is : " + product_id);
        System.out.println("Product name is : " + product_name);
        System.out.println("Price of the product is : " + price);
        System.out.println();
    }
}

public class product {
    public static void main(String[] args) {

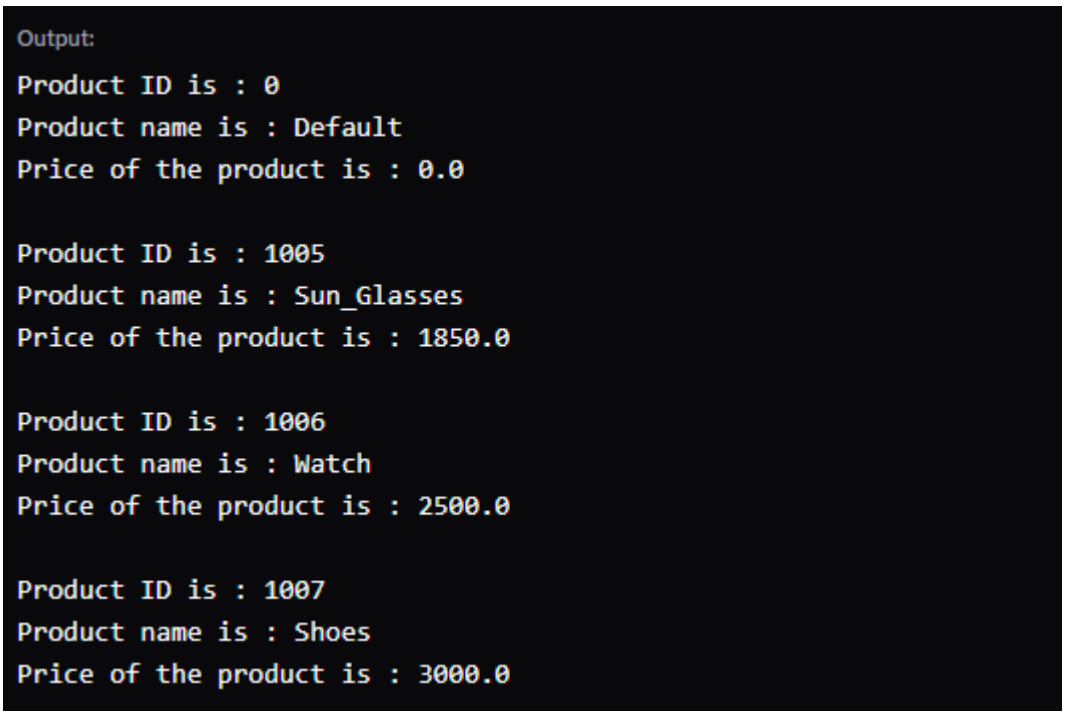
        ArrayList<Product> products = new ArrayList<>();

        Product p1 = new Product();
        Product p2 = new Product(1005, "Sun_Glasses", 1850.00);
        Product p3 = new Product(1006, "Watch", 2500.00);
        Product p4 = new Product(1007, "Shoes", 3000.00);

        products.add(p1);
        products.add(p2);
        products.add(p3);
        products.add(p4);
```

```
        for (Product p : products) {
            p.display();
        }
    }
}
```

Output :



Conclusion: This program demonstrates the use of classes, constructors, ArrayList, and objects in Java. It creates multiple `Product` objects with different details and stores them in an `ArrayList`. The `display()` method is used to show the product information. It also demonstrates constructor overloading by using both a default constructor and a parameterized constructor.